

2、手持激光雷达建图

该功能需要启动slam_gmapping的功能包里的程序，该源码位于【yahboomcar_ws】源码里边。这里以配套的虚拟机说明如何启动程序，如果要放在自己的主板上的话，把yahboomcar_ws放在根目录下编译。

2.1、启动雷达

终端输入，

```
ros2 launch ydlidar_ros2_driver ydlidar_launch.py
```

2.2、发布静态odom转换

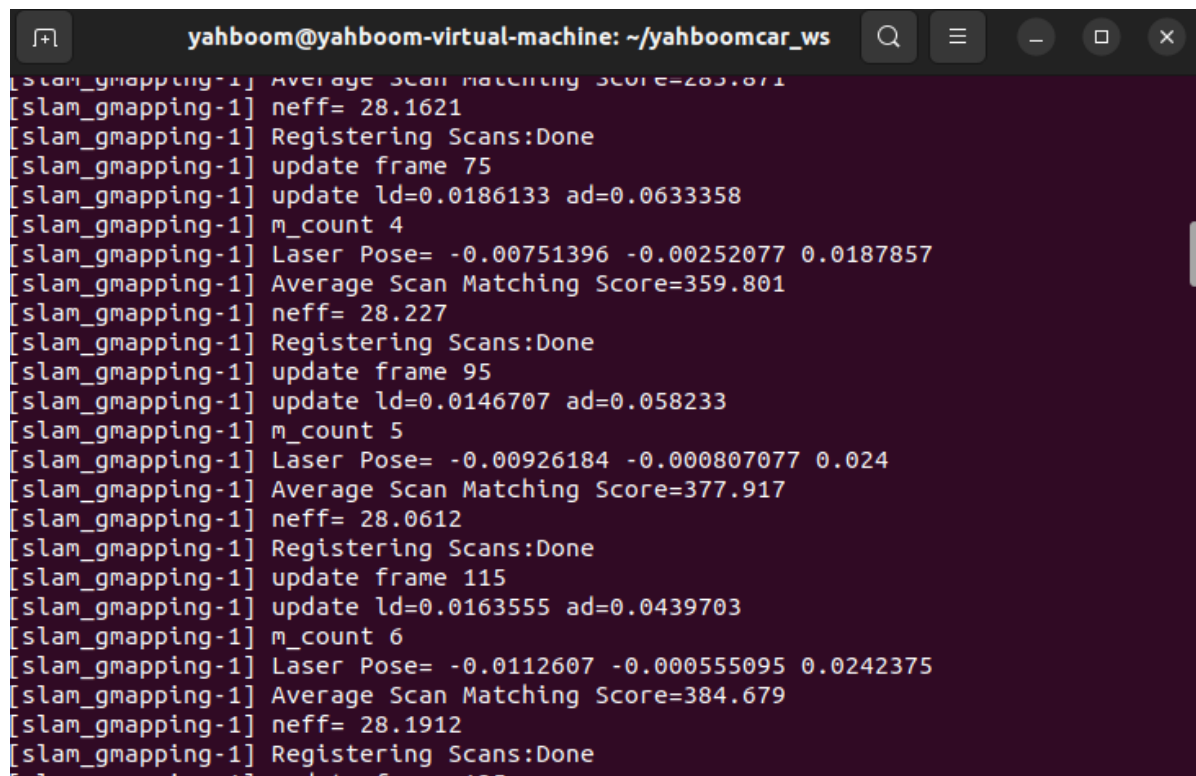
终端输入，

```
ros2 launch rf2o_laser_odometry rf2o_laser_odometry.launch.py
```

2.3、启动gmapping建图

终端输入，

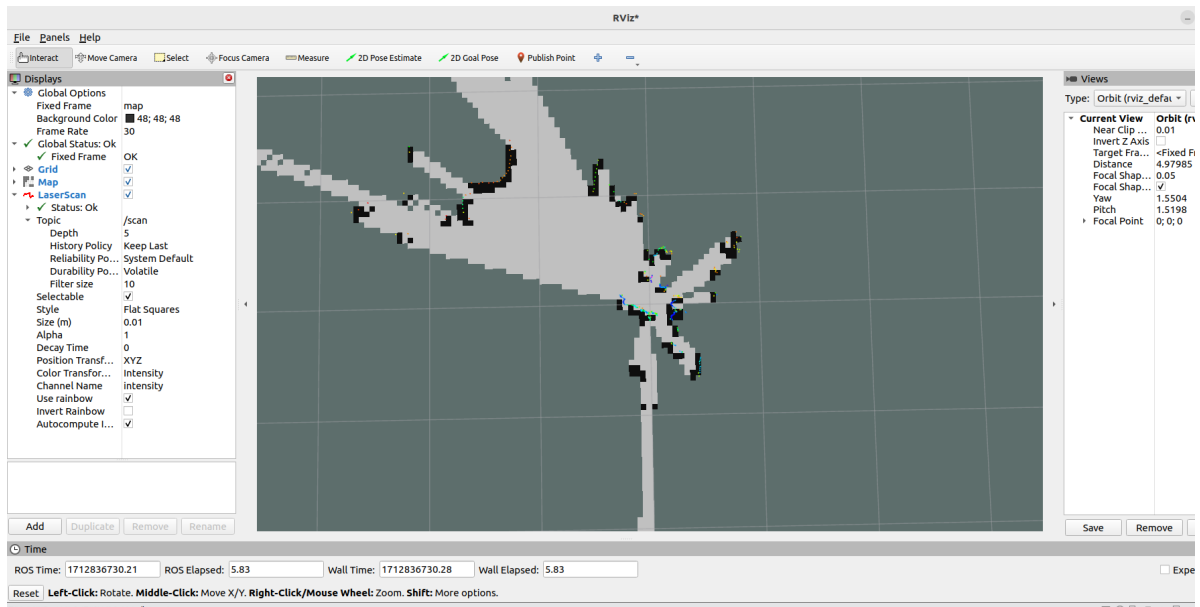
```
ros2 launch slam_gmapping slam_gmapping.launch.py
```

A terminal window titled 'yahboom@yahboom-virtual-machine: ~/yahboomcar_ws' displays the output of the 'slam_gmapping' launch script. The output shows a series of log messages from the 'slam_gmapping-1' process, including 'Average Scan Matching Score', 'neff' values, 'Registering Scans:Done', 'update frame', 'update ld', 'ad', 'm_count', and 'Laser Pose' coordinates. The scores and neff values fluctuate slightly over time, while the laser pose coordinates show small increments in the x, y, and z dimensions.

```
yahboom@yahboom-virtual-machine: ~/yahboomcar_ws
[slam_gmapping-1] Average Scan Matching Score=285.871
[slam_gmapping-1] neff= 28.1621
[slam_gmapping-1] Registering Scans:Done
[slam_gmapping-1] update frame 75
[slam_gmapping-1] update ld=0.0186133 ad=0.0633358
[slam_gmapping-1] m_count 4
[slam_gmapping-1] Laser Pose= -0.00751396 -0.00252077 0.0187857
[slam_gmapping-1] Average Scan Matching Score=359.801
[slam_gmapping-1] neff= 28.227
[slam_gmapping-1] Registering Scans:Done
[slam_gmapping-1] update frame 95
[slam_gmapping-1] update ld=0.0146707 ad=0.058233
[slam_gmapping-1] m_count 5
[slam_gmapping-1] Laser Pose= -0.00926184 -0.000807077 0.024
[slam_gmapping-1] Average Scan Matching Score=377.917
[slam_gmapping-1] neff= 28.0612
[slam_gmapping-1] Registering Scans:Done
[slam_gmapping-1] update frame 115
[slam_gmapping-1] update ld=0.0163555 ad=0.0439703
[slam_gmapping-1] m_count 6
[slam_gmapping-1] Laser Pose= -0.0112607 -0.000555095 0.0242375
[slam_gmapping-1] Average Scan Matching Score=384.679
[slam_gmapping-1] neff= 28.1912
[slam_gmapping-1] Registering Scans:Done
[slam_gmapping-1] update frame 135
```

rviz显示如下，

若没有地图显示，点击左上角的 file --> Open config --> sunrise --> yahboomcar_ws --> src --> slam_gmapping --> rviz --> map_view.rviz 打开rviz地图显示配置



2.4、查看TF树

终端输入，

```
ros2 run tf2_tools view_frames
```

```
yanboom@yanboom-virtual-machine:~/yanboomcar_ws$ ros2 run tf2_tools view_frames
[INFO] [1712836863.067737751] [view_frames]: Listening to tf data for 5.0 second
S...
[INFO] [1712836868.112822280] [view_frames]: Generating graph in frames.pdf file
...
[INFO] [1712836868.118691540] [view_frames]: Result:tf2_msgs.srv.FrameGraph_Response(frame_yaml="odom: \n parent: 'map'\n broadcaster: 'default_authority'\n
rate: 20.197\n most_recent_transform: 1712836868.110051\n oldest_transform: 1712836863.059761\n buffer_length: 5.050\nlaser_frame: \n parent: 'base_link'\n
broadcaster: 'default_authority'\n rate: 10000.000\n most_recent_transform: 0.000000\n oldest_transform: 0.000000\n buffer_length: 0.000\nbase_link: \n pa
rent: 'base_footprint'\n broadcaster: 'default_authority'\n rate: 10000.000\n most_recent_transform: 0.000000\n oldest_transform: 0.000000\n buffer_length:
0.000\nbase_footprint: \n parent: 'odom'\n broadcaster: 'default_authority'\n
rate: 10.237\n most_recent_transform: 1712836867.890208\n oldest_transform: 1712836863.005946\n buffer_length: 4.884\n")
yanboom@yanboom-virtual-machine:~/yanboomcar_ws$
```

会在启动该命令终端的目录下，生成一个frames.pdf文件，这就是生成的TF树，

view_frames Result

Recorded at time: 1712836868.1316395

